PoML Exercise session 3

Learning theory and practical aspects in ML

Exercise types

- Drawing
  - Exercise 1: Applying RPNI
  - Exercise 2: VC dimensions of rectangles
- Calculation
  - Exercise 3: the pizza problem (On the board)
  - Exercise 4: Upper and lower bounds on VS dimension
- Intuition and reasoning:
  - Exercise 5: Selecting the most informative features
  - Exercise 6: Feature selection and sport sciences
- Coding
  - Exercise 7: Coding feature selection methods
  - Exercise 8: Model choice

On VC dimension

On VC dimension

The definition

- The VC-dimension (or Vapnik-Chervonenkis dimension) of a hypothesis space H containing functions $X \rightarrow 0,1$ is the size of the largest finite subset of X that is shattered by H, or infinite if shattered subsets exists for any size.
- Oof.

On VC dimension

What is shattering?

- A set $S \subset X$ of points is shattered by a hypothesis space H if, for each possible label assignment $l : S \to \{0,1\}$ to these points, a hypothesis $h \in H$ exists such that $\forall x \in S : h(x) = l(x)$
- Oof again.

On VC dimension

What is shattering?

- A set $S \subset X$ of points is shattered by a hypothesis space H if, for each possible label assignment $l : S \to \{0,1\}$ to these points, a hypothesis $h \in H$ exists such that $\forall x \in S : h(x) = l(x)$
- X are data points. $S \subset X$ is any selection of those points.
    - For example: $X = \{1,2,3,....\}$ and $S = \{3,5\}$.
- Hypothesis space H is a family of binary classifiers
    - So: they take points in X, and predict 0 or 1
    - For example: $H = \begin{cases} 1 & \text{if } h(x) > a \\ 0 & \text{if } h(x) \leq a \end{cases}$ for all real-valued a
    - Or decision trees.
- A label assignment $l : S \to \{0,1\}$: any combination of $0 - 1$ assignment of the points in S
    - For example, for $S = \{3,5\}$: $l(3) = 0,\ l(5) = 1$
- $\forall x \in S : h(x) = l(x)$
    - a hypothesis (i.e. binary classifier) that classifies the label assignment of S perfectly

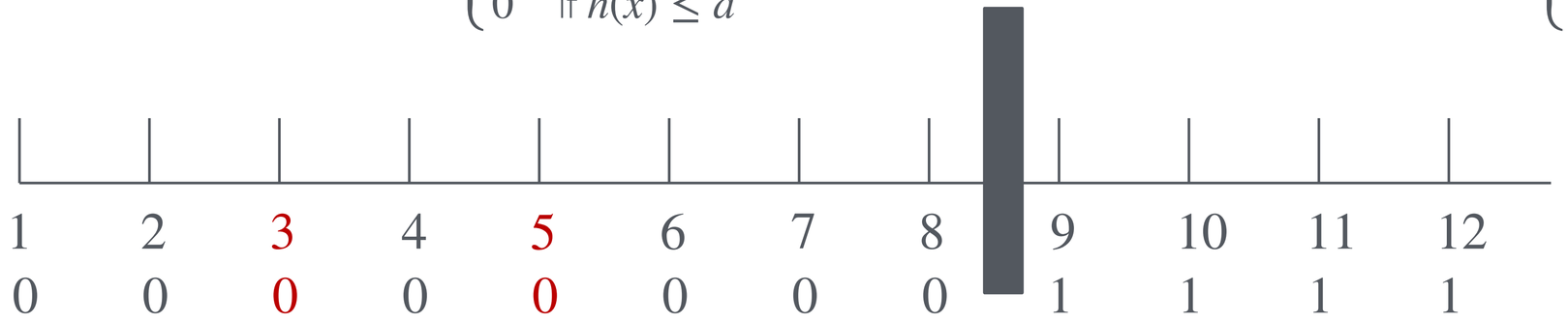
On VC dimension

Shattering: a visual explanation

- Example $X = \{1,2,3,....\}$ with $S = \{3,5\}$:



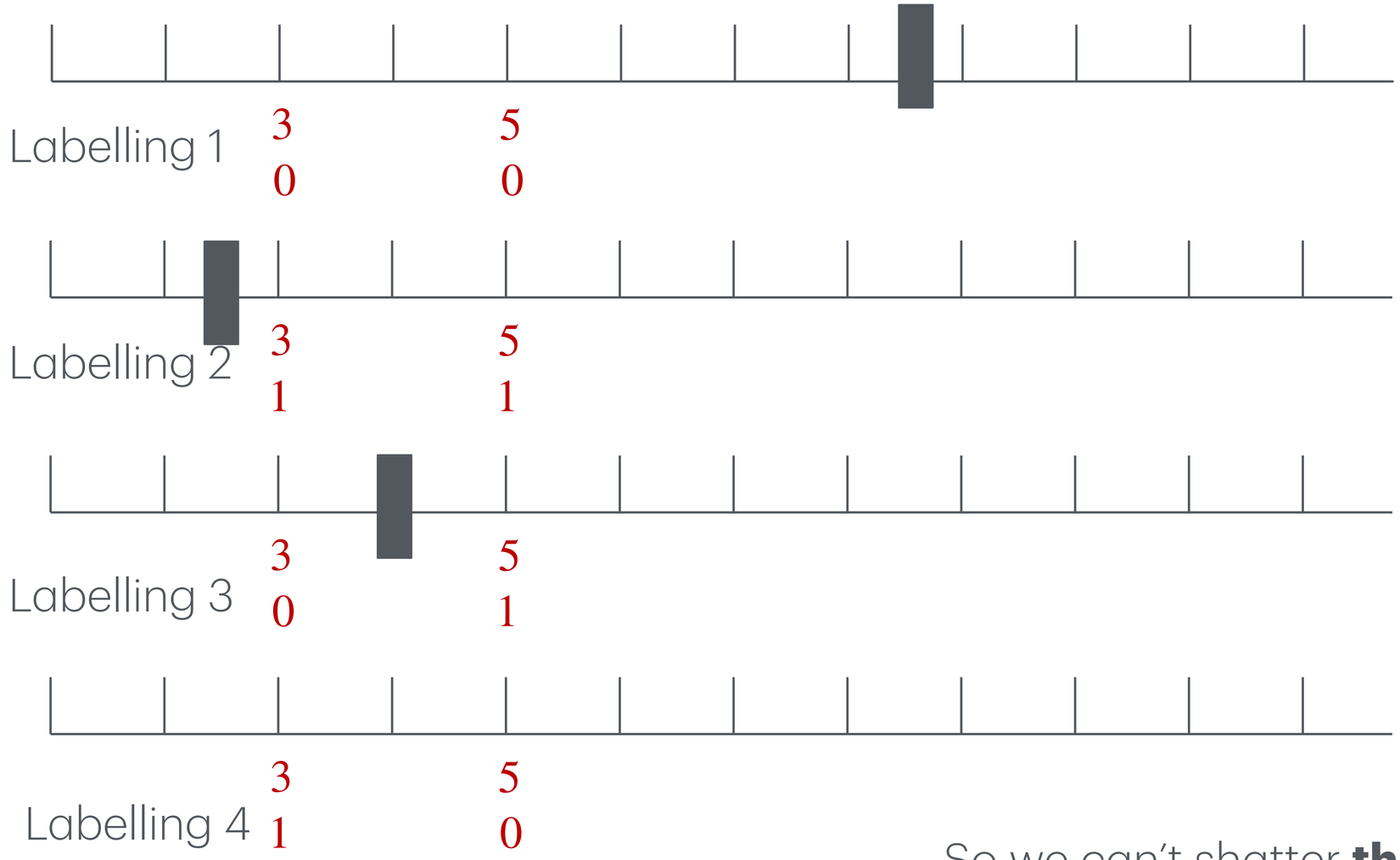
- Hypothesis space $H = \begin{cases} 1 & \text{if } h(x) > a \\ 0 & \text{if } h(x) \leq a \end{cases}$ for all real-valued a. So for example $h = \begin{cases} 1 & \text{if } h(x) > 8.7 \\ 0 & \text{if } h(x) \leq 8.7 \end{cases}$



On VC dimension

Shattering: a visual explanation

S is shattered by H: Find a hypothesis (classifier) for each labelling that can perfectly separate the points.



$$h_1 = \begin{cases} 1 & \text{if } h(x) > 8.7 \\ 0 & \text{if } h(x) \leq 8.7 \end{cases}$$

$$h_2 = \begin{cases} 1 & \text{if } h(x) > 2.5 \\ 0 & \text{if } h(x) \leq 2.5 \end{cases}$$

$$h_3 = \begin{cases} 1 & \text{if } h(x) > 4 \\ 0 & \text{if } h(x) \leq 4 \end{cases}$$

So we can't shatter **this** specific S with H!

On VC dimension

Shattering: a visual explanation

What if $S = \{3\}$?



3

Labelling: 0

$$H = \begin{cases} 1 & \text{if } h(x) > 8.7 \\ 0 & \text{if } h(x) \leq 8.7 \end{cases}$$



3

Labelling: 1

$$H = \begin{cases} 1 & \text{if } h(x) > 2.5 \\ 0 & \text{if } h(x) \leq 2.5 \end{cases}$$

So we **<u>can</u>** shatter **<u>this</u>** specific S with H!

On VC dimension

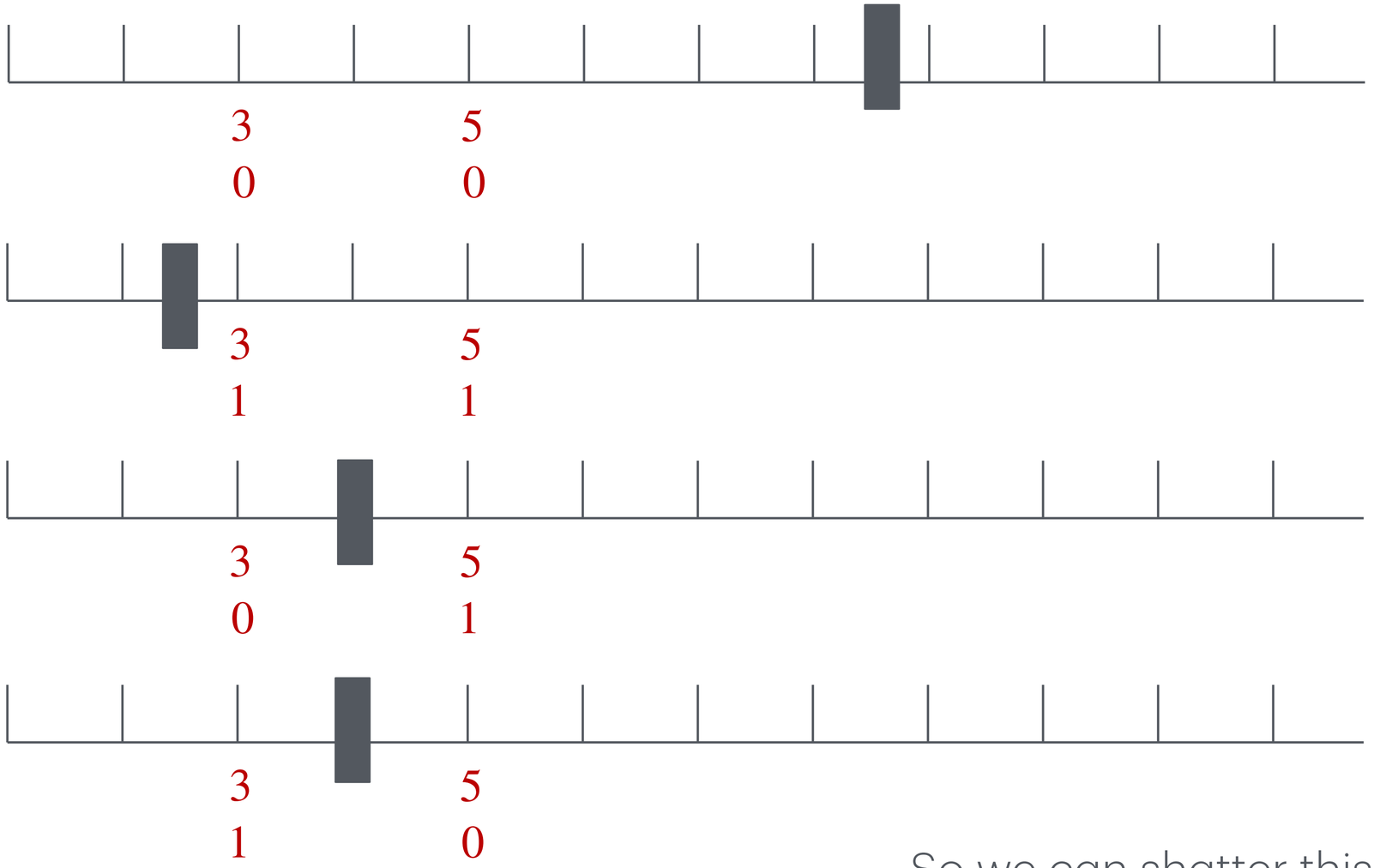
Shattering: a visual explanation

- What if we add Hypothesis space $H' = \begin{cases} 0 & \text{if } h(x) > a \\ 1 & \text{if } h(x) \leq a \end{cases}$ for all real-valued a to H ?



On VC dimension

Shattering: a visual explanation



$$h = \begin{cases} 1 & \text{if } h(x) > 8.7 \\ 0 & \text{if } h(x) \leq 8.7 \end{cases}$$

$$h = \begin{cases} 1 & \text{if } h(x) > 2.5 \\ 0 & \text{if } h(x) \leq 2.5 \end{cases}$$

$$h = \begin{cases} 1 & \text{if } h(x) > 4 \\ 0 & \text{if } h(x) \leq 4 \end{cases}$$

$$h = \begin{cases} 0 & \text{if } h(x) > 4 \\ 1 & \text{if } h(x) \leq 4 \end{cases}$$

So we can shatter this S by $H \cup H'$!

On VC dimension

From shattering to VC dimension

- VC dimension: the size k of the **largest** S that can be shattered by hypotheses in H
  - This could be 1 very specific S
- Why any S of size k? Why not **every** S of size k?
  - There are many S's of size k that will not work.
- Example $X = \{1,2,3,....\}$ with $S = \{5,5\}$



Labelling 0/1

- These 2 points cannot be shattered! Not by H, nor by $H \cup H'$

On VC dimension

Typical exercise 1

- VC dimension: the size k of the **largest** S that can be shattered by hypotheses in H
- A typical exercise: show that the VC dimension of <certain classifiers> over <certain points> is **at least** k
- Step 1: Choose a **smart** set S of k points (Remember: S is fixed!)
- Step 2: write down all possible labellings
- Step 3: for every labelling, find a hypothesis that separates them

On VC dimension

Typical exercise 2

- A typical exercise: show that the VC dimension of <certain classifiers> over <certain points> is **less than** k
- Remember: to show that it is at least k, you need to find 1 S
- So to do that in reverse, you would need to show that it can't be shattered for **every** S
  - This is really difficult!!
- In the slides, you can find formulas for generic upper- and lower bounds.
  - Check for the conditions!

Exercise 3: the pizza problem

Exercise 3

The pizza problem

- Suppose pizzas can be composed of the following 8 ingredients: cheese, ham, tomatoes, mushrooms, peppers, salami, olives and shrimp. Further assume that the concept “pizzas that Wayne likes” can be expressed as a conjunction of “contains I” literals, with I a single ingredient. For instance, “Wayne likes all pizzas that contain peppers and salami” is a possible concept, whereas “Wayne likes all pizzas that contain peppers and no olives” is not. Show that the VC-dimension of this hypothesis space is at least 3 and at most 8



Our X: pizzas

Our labels: ‘Wayne likes <pizza x>

Our H: Wayne likes pizza that adhere to <certain logical formula>

Which logical formulas are in H?

$\wedge$ ✓ $ham \wedge tomatoes$, $ham \wedge peppers \wedge salami$

$\vee$ ✗ $ham \vee tomatoes$, $cheese \wedge (ham \vee tomatoes)$

$\neg$ ✗ $\neg ham$, $cheese \wedge tomatoes \wedge \neg mushrooms$

Exercise 3

What do we need to do?

- Show that the VC dimension of these logical formulas over the pizzas is **at least** 3
- Step 1: Choose 3 **smart** pizzas
- Step 2: write down all possible labellings
- Step 3: for every labelling, find a hypothesis that separates them

Exercise 3

Step 1: let's turn it into math

- Represent the pizzas as 8-bit binary vectors.
- Every index corresponds to 1 ingredient:
  $(cheese, ham, tomatoes, mushrooms, peppers, salami, olives, shrimp)$
- So $(0,0,0,1,1,0,0,1)$ is a pizza with $mushrooms, peppers, shrimp$.
- We need to find 3 binary vectors that we can shatter.

$(cheese, ham, tomatoes, mushrooms, peppers, salami, olives, shrimp)$

Exercise 3

Step 2: start small

- We will construct 3 pizzas x, y, z. We will fill in the gaps:

$$x = (_, _, _, _, _, _, _, _)$$
$$y = (_, _, _, _, _, _, _, _)$$
$$z = (_, _, _, _, _, _, _, _)$$

- Let's say Wayne likes only x.

- Put 1 ingredient on x, that we will not put on the other pizzas:

$$x = (1, _, _, _, _, _, _, _)$$
$$y = (0, _, _, _, _, _, _, _)$$
$$z = (0, _, _, _, _, _, _, _)$$

Our hypothesis: "Wayne likes pizzas with $cheese$"

$(cheese, ham, tomatoes, mushrooms, peppers, salami, olives, shrimp)$

Exercise 3

Step 2: start small

- Let's say Wayne likes only y.
- Put 1 ingredient on y, that we will not put on the other pizzas:

$$x = (1,0,_,_,_,_,_,_)$$
$$y = (0,1,_,_,_,_,_,_)$$
$$z = (0,0,_,_,_,_,_,_)$$

Our hypothesis: "Wayne likes pizzas with ham"

- We have now covered 2 labellings!
- Labelling 1: Wayne likes x. Hypothesis: Wayne likes pizzas with $cheese$
- Labelling 2: Wayne like y. Hypothesis: Wayne likes pizzas with ham.

$(cheese, ham, tomatoes, mushrooms, peppers, salami, olives, shrimp)$

Exercise 3

Step 2: start small

- Labelling 1: Wayne likes x. Hypothesis: Wayne likes pizzas with $cheese$
- Labelling 2: Wayne like y. Hypothesis: Wayne likes pizzas with ham.
- Labelling 3: Wayne like z: Hypothesis: Wayne likes pizzas with $tomatoes$.

$$x = (1,0,0,_,_,_,_,_)$$
$$y = (0,1,0,_,_,_,_,_)$$
$$z = (0,0,1,_,_,_,_,_)$$

$(cheese, ham, tomatoes, mushrooms, peppers, salami, olives, shrimp)$

Exercise 3

Step 3: extend to more labellings

- Labelling 4: Wayne likes x and y.
- Remember: $cheese \vee ham$ is not allowed in our logical formulas
- We put a 4th ingredient on pizza x and y, but not on z
- Hypothesis for labelling 4: Wayne likes pizzas with $mushrooms$

$x = (1,0,0,_,_,_,_,_)$

$y = (0,1,0,_,_,_,_,_)$

$z = (0,0,1,_,_,_,_,_)$

$x = (1,0,0,1,_,_,_,_)$

$y = (0,1,0,1,_,_,_,_)$

$z = (0,0,1,0,_,_,_,_)$

$(cheese, ham, tomatoes, mushrooms, peppers, salami, olives, shrimp)$

Exercise 3

Step 3: extend to more labellings

- Labelling 1: Wayne likes x. Hypothesis: Wayne likes pizzas with $cheese$
- Labelling 2: Wayne like y. Hypothesis: Wayne likes pizzas with ham.
- Labelling 3: Wayne like z: Hypothesis: Wayne likes pizzas with $tomatoes$.
- Labelling 4: Wayne likes x and y. Hypothesis: Wayne likes pizzas with $mushrooms$.
- Labelling 5: Wayne likes x and z. Hypothesis: Wayne likes pizzas with $peppers$.
- Labelling 6: Wayne likes y and z. Hypothesis: Wayne likes pizzas with $salami$.

$$x = (1,0,0,_,_,_,_,_)$$
$$y = (0,1,0,_,_,_,_,_)$$
$$z = (0,0,1,_,_,_,_,_)$$

$$x = (1,0,0,1,1,0,_,_)$$
$$y = (0,1,0,1,0,1,_,_)$$
$$z = (0,0,1,0,1,1,_,_)$$

$(cheese, ham, tomatoes, mushrooms, peppers, salami, olives, shrimp)$

Exercise 3

Step 4: finalize the last cases

- Labelling 1: Wayne likes x. Hypothesis: Wayne likes pizzas with $cheese$

$$x = (1,0,0,1,1,0,_,_)$$

- Labelling 2: Wayne like y. Hypothesis: Wayne likes pizzas with ham.

$$y = (0,1,0,1,0,1,_,_)$$

$$z = (0,0,1,0,1,1,_,_)$$

- Labelling 3: Wayne like z: Hypothesis: Wayne likes pizzas with $tomatoes$.

- Labelling 4: Wayne likes x and y. Hypothesis: Wayne likes pizzas with $mushrooms$.

$$x = (1,0,0,1,1,0,1,_)$$

- Labelling 5: Wayne likes x and z. Hypothesis: Wayne likes pizzas with $peppers$.

$$y = (0,1,0,1,0,1,1,_)$$

$$z = (0,0,1,0,1,1,1,_)$$

- Labelling 6: Wayne likes y and z. Hypothesis: Wayne likes pizzas with $salami$.

- Labelling 7: Wayne likes x,y and z. Hypothesis: Wayne likes pizzas with $olives$.

$(cheese, ham, tomatoes, mushrooms, peppers, salami, olives, shrimp)$

Exercise 3

Step 4: finalize the last cases

A hypothesis for each labelling

8 labellings

- Labelling 1: Wayne likes x. Hypothesis: Wayne likes pizzas with ***cheese***
- Labelling 2: Wayne like y. Hypothesis: Wayne likes pizzas with ***ham***.
- Labelling 3: Wayne like z: Hypothesis: Wayne likes pizzas with ***tomatoes***.
- Labelling 4: Wayne likes x and y. Hypothesis: Wayne likes pizzas with ***mushrooms***.
- Labelling 5: Wayne likes x and z. Hypothesis: Wayne likes pizzas with ***peppers***.
- Labelling 6: Wayne likes y and z. Hypothesis: Wayne likes pizzas with ***salami***.
- Labelling 7: Wayne likes x,y and z. Hypothesis: Wayne likes pizzas with ***olives***.
- Labelling 8: Wayne likes none of x,y and z. Hypothesis: Wayne likes pizzas with ***shrimp***.

$x = (1,0,0,1,1,0,1,_)$

$y = (0,1,0,1,0,1,1,_)$

$z = (0,0,1,0,1,1,1,_)$

3 pizzas

$x = (1,0,0,1,1,0,1,0)$

$y = (0,1,0,1,0,1,1,0)$

$z = (0,0,1,0,1,1,1,0)$

Exercise 3

The final stretch

- The hard part is over. We showed that the VC-dimension of this hypothesis space is at least 3. Now we show that it's at most 8.
- From the slides: $VC(H) \leq \log_2(|H|)$
- $|H| = 2^8$ (either choose an ingredient or not)
- $VC(H) \leq \log_2(2^8) = 8$